MIL-E-1/1044C(USAF)
27 March 1978
SUPERSEDING
MIL-E-1/1044B(USAF)
27 April 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 6263A

This specification is approved for use by the Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pencil triode, UHF, High altitude, glass and metal construction
See figure 1
Mounting position : Any
Weight: 0.85 ounce nominal

ABSOLUTE MAXIMUM RATINGS:

Parameter:	E_f	E_b	E_c	I_b	I_c	I_k	P_p	P_i
Units:	V	Vdc	Vdc	mAdc	mAdc	mAdc	W	W
C. Teleg (note 10)	6.0 ±10%	330	-100	40	25	55	8	13.2
C. Teleg (note 11)	6.0 ±10%	400	-100	55	25	70	13	22
Freq Multiplier (note 10)	6.0 ±10%	300	-125	35	25	45	6	9.9
Freq Multiplier (note 11)	6.0 ±10%	350 (Note 1)	-140	45	25	55	9.5	15.9 (Note 2)

ABSOLUTE MAXIMUM RATINGS:

Parameter:	E_{hk}	R_g	t_k	T(anode seal)	Barometric Pressure Reduced
Units:	Vdc	Neg	sec	Temp °C	mmHg
C. Teleg (note 10)	±50	0.1	60	175	55
C. Teleg (note 11)	±50	0.1	60	175	55
Freq Multiplier (note 10)	±50	0.1	60	175	55
Freq Multiplier (note 11)	±50 (note 3)	0.1	60 min	175	55

TEST CONDITIONS:

Parameter	E_f	E_b	R_k	t_k	E_c
Units:	V	Vdc	ohm	sec	Vdc
	6.0	200	100 (note 4)	180	--- (Note 5)

GENERAL:

Qualification: Required

© Denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1	See notes 6 and 7						
----	Visual and mechanical inspection criteria	See note 8	---	---	---	---	---	---
1301	Heater current		0.65	II	If	265	295	mA
1256	Electrode current (1) (anode)		0.65	II	Ib(1)	18	36	mAdc
1306	Transconductance		0.65	II	Sm	5600	8400	umhos
1266	Total grid current	Rk = 0; Ec = -2.5 Vdc; Rg = 0.5 Meg	0.65	II	Ic	---	-0.5	µAdc
1236	Power oscillation	Eb = 350 Vdc Rg/Ib = 50 mAdc, max Rk = 0; F = 500 ±15 MHz	0.65	II	Po	6.5	---	W useful
		See notes 9 and 10	0.65	II	ΔPo/Ef	---	0.5	W
(C) 1201	Tap shorts	See Note 11	0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1336	Heater cathode leakage	Ehk = +50 Vdc Ehk = -50 Vdc			+Ihk -Ihk	--- ---	100 100	µAdc µAdc
1256	Electrode current (2) (anode)	Rk = 0; Ec = -20 Vdc			Ib(2)	---	55	µAdc
1316	Amplification factor				Mu	20	34	----
(C) 1331	Direct-interelectrode capacitance	See Note 26			Cgp Cgk	1.45 2.45	1.95 3.35	pF pF
1256	Electrode current (dynamic cathode)	Ef = 4.7 V; ec = eb = 25 V anode tied to grid 60 Hz			ik	150	---	mA
1031	Low-frequency vibration	Rk = 0; Ec = -3.0 Vdc; Rp = 10,000 ohms; Vibrate perpendicular to the tube axis			Ep	---	100	mVac
	Quality conformance inspection, part 3							
(C) 1501	Intermittent life	Group A Rg/Ib = 40±2 mAdc; Ebb = 400 Vdc; Ehk = +50 Vdc; Rk = 0; F = 500 ±15 MHz; Rb = 100 ±5 ohms; T(anode seal) See Note 2 tk = 60 sec. min.; T = 500 hours	---	---	---	---	---	---
----	Intermittent life end points:	See notes 13 and 14	---	---	---	---	---	---
----	Inoperatives	See note 15	---	---	Inop.	---	---	---
1236	Power oscillation	See note 16	---	---	Po	5.0	---	W useful
1506	Heater cycling life	Ec = Eb = Ehk = 0 See Notes 17 and 18	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
©	Quality conformance inspection, part 3							
----	Heater cycling life end points:							
© 1211	Insulation of electrodes (R_g-all)	E = 100 Vdc (grid negative)	6.5	S3	R_g-all	50,000	---	ohms
© 1211	Insulation of electrodes (R_p-all)	E = 300 Vdc (anode negative)	6.5	S3	R_p-all	25	---	Meg
1336	Heater-cathode leakage	Ehk = +50 V	6.5	S3	+Ihk	---	150	µAdc
----	Open heaters		6.5	S3	---	---	---	---
© 1516	Stability life	Eb = Vary for: Pp = 2.5 watts min; Pp = 3 watts max; TA = room; Ec = 0; Rk = 0; See Note 19	---	---	----	---	---	---
----	Stability life end point:							
1306B	Change in transconductance of individual tubes	See Note 19	---	---	ΔSmt	---	15	%
1251	Survival-rate life (50 hours)	Stability life test conditions or equivalent, See Note 20	---	---	---	---	---	---
----	Survival rate life end points:	See Notes 20 and 21	---	---	---	---	---	---
----	Inoperatives	See Note 15	1.0	II	Inop.	---	---	---
© 1256	Electrode current (2) (anode)		1.0	II	Ib(2)	---	100	µAdc
1236	Power oscillation		1.0	II	Po	5.0	---	W useful
© 1002	Barometric pressure, reduced	Pressure = 55 +5mmHg; Voltage = 500 Vac See Notes 22 and 23	---	--	---	---	---	---
© ----	Seal fracture	No voltages See Notes 23 and 24	---	---	---	60	---	lbs.
© 1031	High-frequency vibration	No voltages; F = 40-60 Hz; t = 180 sec; Accel = 10 G; Vibrate perpendicular to the tube axis See Notes 23 and 25	---	---	---	---	---	---
© 1211	Electrode insulation	Ef = 6.0 V; E = 100 Vdc (grid negative)	---	---	R_g-all	25	---	Meg
		E = 300 Vdc (anode negative) See Note 23	---	---	R_p-all	25	---	Meg
© 1331	Direct-interelectrode capacitance	See Notes 23 and 26	---	---	cpk	---	0.08	pF

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NOTES:

1. For tube protection, it is essential that sufficient DC resistance be used in the plate supply circuit so that the short circuit current is limited to two amperes.

2. If Po exceeds 2.5 watts, sufficient air must be directed onto the anode radiator to maintain the anode glass to metal seal temperature at, or less than, 175°C.

3. The reference point for heater-cathode potential shall be the positive terminal of the cathode resistor.

4. Reactance of the cathode by-pass capacitor shall be less than 3 ohms.

5. Whenever Ec is not specified on any test the grid shall be grounded directly or through a grid resistor (if specified) except where related MIL-STD-1311 test methods imply grid variance to accomplish the test.

6. Use single, double, or multiple sampling provisions per MIL-STD-105.

7. Characteristics within this group shall be sampled in accordance with MIL-STD-105, inspection level II, and shall, excluding inoperatives and mechanical, meet a combined AQL of 1 percent. The AQL for individual characteristics is 0.65 percent. A lot failing the 1 percent combined AQL because of defects for one characteristic may be detailed for that characteristic alone. However, if the lot failure results from defects of more than one characteristic, detailing shall include all characteristics in the group.

8. This inspection shall be performed visually without the aid of magnification. These tests shall be classified as production tests and shall be conducted at inspection level I of MIL-STD-105, with the following AQL values:

Type of Defect	AQL
Major 1	1 percent (combined defectives)
Major 2	1 percent (each defect)
Minor	2.5 percent (combined defectives)
Control	6.5 percent (each defect)

8.1 Plating

8.1.1	There shall be no scratches exposing base metal exceeding 1/4 inch in length or more than 0.003 inch wide.	Minor
8.1.2	Only one scratch per plated part is allowable.	Minor
8.1.3	Scratches not exposing base metal are acceptable.	Minor
8.1.4	There shall be no exposed base metal exceeding 1/3 of the edge of the cathode terminal.	Control
8.1.5	Peeling or Marks	
8.1.5.1	There shall be no evidence of peeling plating.	Major 1
8.1.5.2	The total area of blisters shall not exceed 0.0010 square inch.	Control
8.1.5.3	The total area of surface abrasions and pinpoints or marks shall not exceed 0.0010 square inch.	Control
8.1.5.4	No individual item in 8.1.5.2 or 8.1.5.3 shall exceed 0.0005 square inch.	Control

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NOTES: -Continued

8. Cont'd

8.2	Coated leads	
8.2.1	The uncoated lead length and adjacent to the glass stem shall not exceed 0.080 inch maximum.	Major 1
8.2.2	The pin diameter shall not exceed plus 0.002 inch from the bogey lead diameter.	Control
8.2.3	There shall be no evidence of solder splash or flux residue on the stem seal surface.	Minor
8.2.4	Criticize if bare spot encircles lead or if any spot is 0.04 inch long or greater except as covered in 8.2.1.	Control
8.3	Radiator	
8.3.1	No fins shall be bend more than 1/2 the distance between fins.	Control
8.3.2	There shall be no solder on the main radiator fin.	Minor
8.3.2	There shall be no copper color through the plating of fin adjacent to anode seal; copper color on other fins is not cause for rejection.	Minor
8.4	Rust	
8.4.1	There shall be no evidence of rust on the ferrous parts of the tube.	Major 1
8.5	General	
8.5.1	Tarnish is not rejectable.	-----
8.5.2	There shall be no exhaust tublation solder or silver meltings that visually increase the anode terminal diameter.	Minor
8.5.3	There shall be no glass from stem sealing on the cathode terminal surface. The outer edge of the glass button shall not protrude radially beyond the extended cylindrical surface of the cathode terminal.	Minor
8.5.4	There shall be no total burned area in excess of 0.0025 square inch at the exhaust tubulation end of the anode.	Control
8.5.5	External surfaces of the ferrous parts shall be silver or gold plated 50 milligrams/square inch minimum.	-----

9. Adjust Rg for the specified Ib = 50 mAdc maximum, only during the first 10 seconds of the test period. The feedback setting shall not be adjusted for individual tubes. Record the power output.

10. At the conclusion of the Po test at Ef = 6.0 V, reduce Ef to 5.0 V, allow 30 seconds for the tube to stabilize, with no other adjustments, record the Po test at Ef = 5.0 V. Reject the tube for slump if the Po change from Ef = 6.0 V to Ef = 5.0 V exceeds 0.5 watts.

©11. Tubes shall be tested in accordance with Method 1201, except that the tube shall be placed in an adaptor and mounted on a paddle (267-JAN or equivalent). The paddle shall be tapped with a rubber tapper three times in each of two planes, 90° to 120° apart.

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NOTES: -Continued

13. The average life of the life test sample shall not be less than 80 percent for combined power oscillation and inoperatives. The definition of an inoperative shall be in accordance with Note 15.

14. The average life of the sample shall be calculated as follows:

    a. If the tube successfully completes the numbers of hours specified, the number of credit hours shall be considered equal to the specified duration of the test.

    b. If the time of failure of a tube was determined exactly, the number of credit hours shall be equal to the number of hours on life test completed before failure.

    c. If the time of failure of a tube cannot be determined exactly, the number of credit hours shall be computed by one of the following methods, whichever yields the lesser results.

        (1) The sum of the hours earned up to the time of the last successful reading plus 10 percent of the life-test duration specified.

        (2) The number of hours midway between the time of the last successful reading to the time or the first unsuccessful reading.

    The average life of the sample shall be the average of the hours credited to the individual tubes in that sample. Unless otherwise specified, at the conclusion of the time specified for the life test, the average life of the sample shall be not less than 80 percent of the test.

15. An inoperative shall be defined as a tube having one or more of the following defects:

    Discontinuity (1201)
    Permanent shorts (1201)
    Tap shorts (1521) survival rate life test only
    Air leak, grid current exceeding 1.0 μa (1267)
    Power oscillation less than 2.0 watts

16. Use power oscillation test conditions. At zero hours, record Rg value for Ib = 50 mAdc on each tube under test. At subsequent reading periods, and at end of life, set Rg at initial zero hour value and record power output.

17. The tubes shall have the filament voltage cycled at a duty cycle of 3 minutes "ON" and 3 minutes "OFF." This shall be a destructive test. Use single sampling; normal, tightened, or reduced inspection, in accordance with MIL-STD-105.

© 18. Test may be performed on a Group D basis every 90 days with the lot consisting of the current month and two previous months. Failure of any tube shall revert back to Group D monthly lot sampling. After three consecutive successful lot submissions have been completed, the test may revert to the quarterly basis.

19. Stability Life Test

    a. Serially mark all tubes from the sample.
    b. Record referenced characteristic measurement.
    c. Operate for 1 hour (plus 30 minutes minus 0 minutes).
    d. Record referenced characteristic measurements at the end of this life test period.
    e. A defective shall be defined as a tube having a change in referenced characteristic greater than specified on the tube specification sheet.

$$\text{Change in percentage referenced characteristic} = \frac{\text{initial value minus one hour value}}{\text{initial value}} \times 100$$

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NOTES: -Continued

20. Survival rate life test

a. Operate survival rate life test sample for 50 hours under conditions specified.
b. Tube shall be tested at the end of this period for the characteristic specified. MIL-STD-105, inspection level II to apply.
c. Determine the number of defective tubes at the 50-hour period.
d. If more than the allowable number of defectives occur, declare the lot nonconforming.
e. An inoperative for this test shall be defined as a tube having one or more of the defects listed in Note 15.
f. At the end of 50 hours, tubes which meet the initial test requirements shall not be considered to have undergone the destructive test.

21. Order of evaluation life test defects: If a tube is defective for more than one attribute characteristic, the characteristic appearing first in the life-test end points shall constitute the failure.

22. The specified voltage shall be applied between the anode and the grid plus cathode. Tubes that show evidence of corona or arcing shall be considered defective. This is a nondestructive test.

©23. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of Quality Conformance Inspection, Part 2, code level D, with an AQL of 6.5.

24. Tube supported at cathode and at a point between radiator fins and anode flange with the grid disc centered. The supports shall be spaced 15/16 ±1/64 inch apart. Force applied to inner grid disc without shock, perpendicular to axis of tube.

25. Tubes shall meet post vibration test: +Ihk, low frequency vibration, electrode current (2).

©26 A capacitance bridge operating frequency of approximately 1 to 2 kHz may be substituted in lieu of approximately 500 kHz.

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Preparing activity:
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Review activity:
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LTR NOTE 9	DIMENSIONS IN INCHES METRIC EQUIVALENTS IN ( )	
	MIN	MAX
QUALITY CONFORMANCE INSP. PART 1		
J	.535 (13.59)	.575 (14.61)
K	.245 (6.22)	.255 (6.48)
N	.808 (20.52)	.816 (20.73)
S	.815 (20.70)	.865 (21.97)
Z	–	.035 (0.89)
QUALITY CONFORMANCE INSP. PART 2		
A	–	1.010 (25.65)
D	.275 (6.99)	.305 (7.75)
F	.600 (15.24)	–
L	–	.400 (10.16)
M		.400 (10.16)
P	–	2.340 (59.44)
R	.200 (5.08)	.290 (7.37)
U	.095 (2.41)	.135 (3.43)
V	.018 (0.46)	.022 (0.56)
W	.009 (0.23)	.015 (0.38)
QUALITY CONF. INSP. PT. 3 (TYPE APPROVAL)		
B	–	.815 (20.70)
C	.416 (10.57)	.424 (10.77)
G	–	.900 (22.86)
H	.360 (9.14)	.440 (11.18)
Q	–	.035 (0.89)
T	–	.553 (14.05)
X	.060 (1.52)	.090 (2.29)

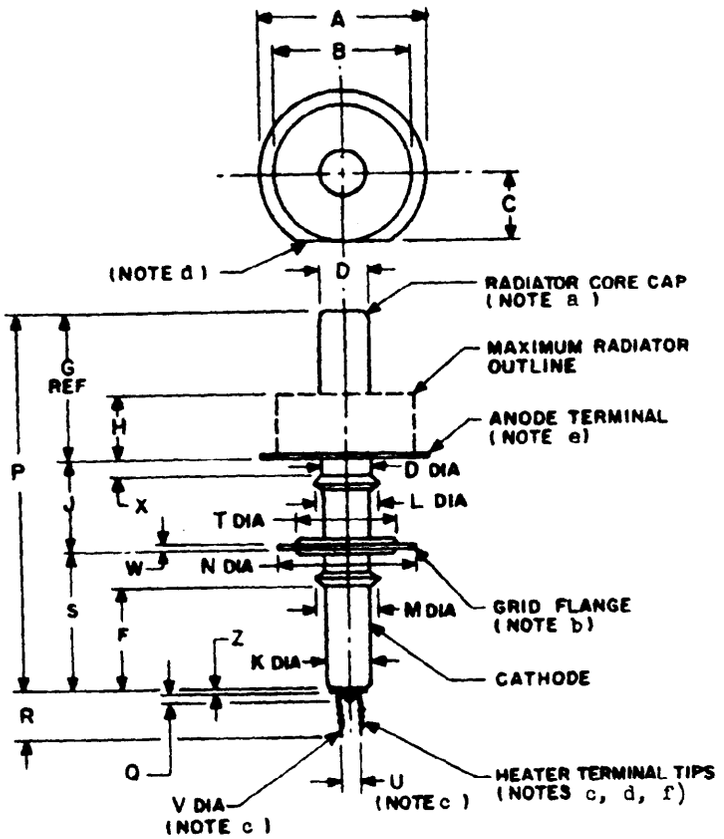


Figure 1. Outline Drawing of Electron Tube Type 6263A.

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NOTES:

a. Eccentricity of radiator and cap shall not exceed 0.030 inch, measured by chucking cathode 0.050 - 0.100 inch from cathode flange and measuring one half total run-out at a point 1/8 inch from the tip.

b. Tilt of grid flange with respect to rotational axis of cathode terminal shall not exceed 0.025 inch. Chuck cathode as specified in note a, and gauge total run-out of flange at a point approximately 0.020 inch inward from its edge for one complete revolution. Eccentricity of grid flange shall not exceed 0.015 inch, as determined by chucking the cathode as specified in note a, and measuring one half total run-out of the grid disk rim for one complete revolution.

c. Tubes shall be 100 percent pin straightened.

d. The flat section on the perimeter of the cavity closure fin shall be parallel to a plane through the heater leads at the glass within 15°

e. Tilt of cavity closure fin of radiator with respect to the rotational axis of cathode terminal as determined by chucking the cathode as specified in note a, and measuring the total travel distance of the cavity closure fin parallel to the axis at a point 0.020 inch inward from the straight edge of the cavity closure fin for one complete revolution (0.035 inch maximum). Radiator closure fin shall be firmly affixed, show no evidence of lateral or rotational movement.

f. Dimension U shall be measured at the outer surface of the glass seal.

g. Sample and criticize in accordance with MIL-STD-105.

Dimension	AQL	Inspection Level
J, K, Note c	1%	Combined I
Z, N, S	2.5%	Combined I
A, D, F, L, M, P, R, U, V, W, Notes, a, b, d, e, f	6.5%	Individual Code letter G

h. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-E-1/1044C (USAF) Electron Tube, Negative Grid (Microwave) Type 6263A FSC 5960

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